Chapter 3: Analysis of LTI System in Frequency Domain

3.1. Frequency response of LTI system, response to complex exponential
3.2. Linear constant co-efficient difference equation and corresponding system function
3.3. Relationship of frequency response to pole-zero of system
3.4. Linear phase of LTI system and its relationship to causality.

Introduction:

In this chapter, we develop the characterization of LTI systems in the frequency domain. The basic excitation of signals in this development are the complex exponentials and sinusoidal functions. The characteristics of the system are described by a function of the frequency variable ω called the frequency response, which is the Fourier transform of the impulse response h[n] of the system.

The System Function of an LTI System:

We know, for an LTI system

$$x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n]$$

$$y[n] = x[n] * h[n]$$

Taking z-transform, Y(z) = X(z)H(z) => H(z) = Y(z)/X(z)

If we know the x[n] and we observe the output y[n] of the system, we can determine the unit sample response. It is clear that H(z) represents the z-domain characterization of a system, whereas h[n] is the corresponding time-domain characterization of the system. The transform function H(z) is called the system function.

When the system is described by an LCCD equation,

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k]$$

Taking z-transform

$$Y(z) = -\sum_{k=1}^{N} a_k z^{-k} Y(z) + \sum_{k=0}^{M} b_k z^{-k} X(z)$$

$$Simplifying\ we\ get, \qquad H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

Hence, and LTI system described by a LCCD equation has a rational system function.

Case I: If $a_k = 0$ for $1 \le k \le N$

$$H(z) = \sum_{k=0}^{M} b_k z^{-k} = \frac{1}{z^M} \sum_{k=0}^{M} b_k z^{M-k}$$

In this case, H(z) contains M zeros, whose values are determined by system parameters $\{b_k\}$ & an M^{th} order pole at z=0 (trivial poles). Hence the system is called all-zero system. Such a system has finite duration impulse response & is called FIR system.

Case II: If $b_k = 0$ for $1 \le k \le M$,

$$H(z) = \frac{b_0}{\sum_{k=1}^{N} a_k z^{-k}} = \frac{b_0 z^N}{\sum_{k=1}^{N} a_k z^{N-k}}, \ a_0 \cong 1$$

In this case H(z) consists of N poles, whose values are determined by the system parameters $\{a_k\}$ & and N^{th} order zero at z = 0. The corresponding system is called all-pole system. Due to presence of poles, the impulse response of such a system is infinite in duration & hence it is an IIR system.

Response of Complex Exponential and Sinusoidal Signals

The response of any relaxed LTI system to an arbitrary input signal x[n] is given by the convolution sum as

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

In this input-output relationship, the system is characterized in the time domain by its unit sample response h[n],

To develop a frequency-domain characterization of the system, let us excite the system with the complex exponential

$$x[n] = Ae^{j\omega n}, \qquad -\infty < n < \infty$$

Where A is the amplitude and ω is any arbitrary frequency confined to the frequency interval [-π, π]. Substituting the value of x[n], we obtain the response

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\left[Ae^{j\omega(n-k)}\right] = A\left[\sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}\right]e^{j\omega n}$$

We obverse that the term in brackets in above equation is a function of the frequency variable ω. In fact, this term is the Fourier transform of the unit sample response h[k] of the system. Hence, we denote this function as

$$H(e^{j\omega}) = H(\omega) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

Clearly, the function H(ω) exists if the system is BIBO stable, that is, if

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Therefore, the response of the system to the complex exponential is given as

$$y[n] = AH(\omega)e^{j\omega n}$$

We note that the response is also in the form of a complex exponential with the same frequency as the input, but altered by the multiplicative factor H(ω).

As a result of this characteristic behavior, the exponential signal is called an *eigenfunction* of the system. In other words, an eigenfunction of a system is an input signal that produces an output that differs from the input by a constant multiplicative factor, H(ω) (called eigenvalue of the system).

 # Determine the output sequence of the system with impulse response

$$\boldsymbol{h[n] = \left(\frac{1}{2}\right)^n u[n]}$$

when the input is the complex exponential sequence

$$\boldsymbol{x[n] = Ae^{j\pi n/2}}, \qquad -\infty < \boldsymbol{n} < \infty$$

Solution: First we evaluate the Fourier transform of the impulse response h[n],

$$H(\omega) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n} = \frac{1}{1-\frac{1}{2}e^{-j\omega}}$$

$$at\ \omega = \frac{\pi}{2}, \qquad H\left(\frac{\pi}{2}\right) = \frac{1}{1-\frac{1}{2}e^{-\frac{j\pi}{2}}} = \frac{1}{1+j\frac{1}{2}} = \frac{2}{\sqrt{5}}e^{-j26.6^0}$$

$$Therefore, the\ output\ is\ (\omega = \pi/2), \qquad y[n] = AH(\pi)e^{j\pi n/2} = A\left(\frac{2}{\sqrt{5}}e^{-j26.6^0}\right)e^{j\pi n/2}$$

$$y[n] = \frac{2}{\sqrt{5}}Ae^{j(\pi n/2 - 26.6^0)}, \qquad -\infty < n < \infty$$

This example clearly illustrates that the only effect of the system on the input signal is to scale the amplitude by $2/\sqrt{5}$ and shift the phase by -26.6^0. Thus, the output is also a complex exponential of frequency $\pi/2$, amplitude $2A/\sqrt{5}$, and phase -26.6^0.

In general, H(ω) is a complex-valued function of the frequency variable ω. Hence it can be expressed in polar form as

$$H(\omega) = |H(\omega)|e^{j\emptyset(\omega)}$$

Where |H(ω)| is the magnitude and φ(ω) is the phase shift imparted on the input by the system at the frequency ω.

For a linear time-invariant system with a real-valued impulse response, the magnitude and phase functions possess symmetry properties. The magnitude |H(ω)| is an even function of ω and the phase φ(ω) is an odd function of ω. Hence if we know |H(ω)| and φ(ω) for $0 \le \omega \le \pi$, we also know these functions for $-\pi \le \omega \le \pi$.

Response of LTI system to sinusoid:

We know, if the input is

$$x_1[n] = Ae^{j\omega n}, \qquad the\ output\ is, \qquad y_1[n] = A|H(\omega)|e^{j\emptyset(\omega)}e^{j\omega n}$$

$$similarly, x_2[n] = Ae^{-j\omega n}, \qquad the\ output\ is, \qquad y_2[n] = A|H(-\omega)|e^{j\emptyset(-\omega)}e^{-j\omega n}$$

Applying symmetry properties

$$y_2[n] = A|H(\omega)|e^{-j\emptyset(\omega)}e^{-j\omega n}$$

Now, by applying the superposition property of the LTI system, we find the response of the system to the input

$$x[n] = \frac{1}{2}\{x_1[n] + x_2[n]\} = A\cos\omega n, \qquad is$$

$$y[n] = \frac{1}{2}\{y_1[n] + y_2[n]\} = A|H(\omega)|\cos[\omega n + \emptyset(\omega)]$$

Similarly, if

$$x[n] = \frac{1}{j2}\{x_1[n] - x_2[n]\} = A\sin\omega n,$$

$$y[n] = \frac{1}{j2}\{y_1[n] - y_2[n]\} = A|H(\omega)|\sin[\omega n + \emptyset(\omega)]$$

It is apparent from this discussion that H(ω) completely characterize the effect of the system on a sinusoidal input signal of any arbitrary frequency. By knowing H(ω), we are able to determine the response of the system to any sinusoidal input signal. Since H(ω) specifies the response of the system in the frequency domain, it is called frequency response of the system.

Correspondingly, $|H(\omega)|$ is called the magnitude response and $\varphi(\omega)$ is called the phase response of the system. If the input to the system consists of more than one sinusoid, the superposition property of the linear system can be used to determine the response.

The Frequency Response Function:

We know that if the system function H(z) converges on the unit circle, we can obtain the frequency response of the system by evaluating H(z) on the unit circle. Thus

$$H(\omega) = H(z)|_{z=e^{j\omega}} = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$$

In the case where H(z) is a rational function of the form H(z)=B(z)/A(z), we have,

$$H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

$$H(z) = b_0 z^{N-M} \left(\prod_{l=1}^{N}(z - z_l)\right) / \left(\prod_{k=1}^{N}(z - p_k)\right)$$

If the ROC of H(z) includes a unit circle ($z = e^{j\omega}$), then we can evaluate H(z) on unit circle, resulting in a frequency response function or transfer function $H(e^{j\omega})$

$$H(e^{j\omega}) = b_0 e^{j(N-M)\omega} \left(\prod_{l=1}^{N}(e^{j\omega} - z_l)\right) / \left(\prod_{k=1}^{N}(e^{j\omega} - p_k)\right)$$

Here magnitude response function,

$$|H(e^{j\omega})| = |b_0| \times \frac{|e^{j\omega} - z_1||e^{j\omega} - z_2| \ldots . |e^{j\omega} - z_M|}{|e^{j\omega} - p_1||e^{j\omega} - p_2| \ldots . |e^{j\omega} - p_N|}$$

The phase response function,

$$Arg\left(H(e^{j\omega})\right) = \{0 \; or \; \pi\} + (N - M)\omega + \sum_{k=1}^{M} Arg(e^{j\omega} - z_k) \; - \sum_{k=1}^{N} Arg(e^{j\omega} - p_k)$$

Evaluate the frequency response of the system described by the system function

$$H(z) = \frac{1}{1 - 0.8z^{-1}} = \frac{z}{z - 0.8}$$

Solution: Clearly, H(z) has a zero at z = 0 and a pole at p =0.8. Hence, the frequency response of the system is

$$H(z)|_{z=e^{j\omega}} = \frac{1}{1 - 0.8e^{-j\omega}}$$

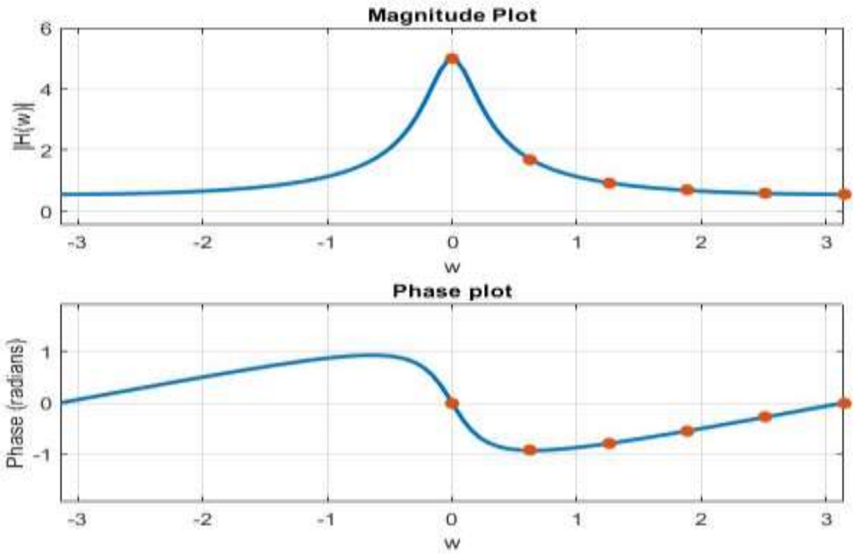
The magnitude response,

$$|H(\omega)| = \left|\frac{1}{1 - 0.8e^{-j\omega}}\right| = \frac{1}{\sqrt{(1 - 0.8cos\omega)^2 + (0.8sin\omega)^2}} = \frac{1}{\sqrt{1.64 - 1.6cos\omega}}$$

And the phase response is

$$\emptyset(\omega) = 0 - \tan^{-1}\left(\frac{0.8sin\omega}{1 - 0.8cos\omega}\right)$$

ω	0	π/5	2π/5	3π/5	4π/5	π
$\lvert H(\omega)\rvert$	5	1.7011	0.9343	0.6845	0.5838	0.5556
$\emptyset(\omega)$	0	-0.9271	-0.7907	-0.5478	-0.2781	-0



Linear phase of LTI system and its relationship to causality.

In the context of Linear Time-Invariant (LTI) systems, the concept of linear phase is crucial for understanding how signals are processed without distortion. Linear phase is characteristic of an ideal filter.

Let us assume that a signal sequence {x[n]} with frequency components confined to the frequency range $\omega_1 < \omega < \omega_2$ is passed through a filter with frequency response

$$H(\omega) = \begin{cases} Ce^{-j\omega n_0}, & \omega_1 < \omega < \omega_2 \\ 0, & otherwise \end{cases}$$

Where C and n_0 are constants. The signal at the output of the filter has a spectrum

$$Y(\omega) = X(\omega)H(\omega) = CX(\omega)e^{-j\omega n_0}, \qquad \omega_1 < \omega < \omega_2$$

By applying the scaling and time-shifting properties of the Fourier transform, we obtain the time-domain output

$$y[n] = Cx[n - n_0]$$

Consequently, the filter output is simply a delayed and amplitude-scaled version of the input signal. A pure delay is usually tolerable and is not considered a distortion o the signal. Neither is amplitude scaling. Therefore, ideal filters have a linear phase characteristic within their passband, that is,

$$\emptyset(\omega) = -\omega n_0$$

The term $-\omega n_0$ indicates a linear phase shift, which means that all frequency components of the input signal are delayed by the same amount of time, preserving the waveform shape.

The derivative of the phase with respect to frequency has the units of delay. Hence, we can define the signal delay as a function of frequency as

$$\tau_g(\omega) = -\frac{d\varphi(\omega)}{d\omega}$$

$\tau_g(\omega)$ is usually called the envelope delay or the group delay of the filter. We interpret $\tau_g(\omega)$ as the time delay that a signal component of frequency ω undergoes as it passes from the input to the output of the system. Note that $\varphi(\omega)$ is linear, $\tau_g(\omega) = n_0$ = constant. In this case all frequency components of the input signal undergo the same time delay.

In conclusion, ideal filters have a constant magnitude characteristic and a linear phase characteristic within their passband. In all cases, such filters are not physically realizable but serve as a mathematical idealization of practical filters.

Relationship to Causality

Causality is a fundamental property of physical systems, meaning that the output of the system at any time depends only on past and present inputs, not future ones. For an LTI system to be causal, its impulse response h[n] must be zero for $n < 0$.

The relationship between linear phase and causality can be summarized as follows:

- **Linear Phase and Symmetry**: For an LTI system to have a linear phase, its impulse response h[n] must be symmetric or anti-symmetric. This symmetry ensures that the phase response is linear.
- **Causality and Symmetry**: A causal system cannot have a symmetric impulse response unless it is non-zero only for $n \geq 0$. This constraint means that achieving both perfect linear phase and causality is challenging.

In practice, achieving an exact linear phase while maintaining causality often requires compromises, such as using finite impulse response (FIR) filters, which can be designed to approximate linear phase while being causal.

Reference:

1. J. G. Proakis, D. G. Manolakis, "Digital Signal Processing, Principles, Algorithms and Applications", 3rd Edition, Prentice-hall, 2000. Chapter 3.